MATHHUB_PRO // NCERT_SOLUTIONS

ANSWERS

Chapter 1

1.1 (a) 10^{-6} ; (b) 1.5×10^{4} ; (c) 5 ; (d) 11.3, 1.13×10^{4}.

1.2 (a) 10^{7} ; (b) 10^{-16} ; (c) 3.9×10^{4} ; (d) 6.67×10^{-8}.

1.5 500

1.6 (c)

1.7 0.035 mm

1.9 94.1

1.10 (a) 1 ; (b) 3 ; (c) 4 ; (d) 4 ; (e) 4 ; (f) 4.

1.11 8.72 m^2; 0.0855 m^3

1.12 (a) 2.3 kg ; (b) 0.02 g

1.13 The correct formula is $m = m_0 (1 - v^2/c^2)^{-½}$

1.14 $\cong 3 \times 10^{-7}$ m^3

1.15 $\cong$ 10^4; intermolecular separation in a gas is much larger than the size of a molecule.

1.16 Near objects make greater angle than distant (far off) objects at the eye of the observer. When you are moving, the angular change is less for distant objects than nearer objects. So, these distant objects seem to move along with you, but the nearer objects in opposite direction.

1.17 1.4×10^{3} kg m^{-3}; the mass density of the Sun is in the range of densities of liquids / solids and *not* gases. This high density arises due to inward gravitational attraction on outer layers due to inner layers of the Sun.

Chapter 2

2.1 (a), (b)

2.2 (a) A....B, (b) A....B, (c) B....A, (d) Same, (e) B....A....once.

2.4 37 s

2.5 3.06 m s^{-2} ; 11.4 s

© NCERT not to be republished

2.6 (a) Vertically downwards; (b) zero velocity, acceleration of 9.8 m s^{-2} downwards; (c) $x > 0$ (upward and downward motion); $v < 0$ (upward), $v > 0$ (downward), $a > 0$ throughout; (d) 44.1 m, 6 s.

2.7 (a) True;, (b) False; (c) True (if the particle rebounds instantly with the same speed, it implies infinite acceleration which is unphysical); (d) False (true only when the chosen positive direction is along the direction of motion)

2.10 (a) 5 km h^{-1}, 5 km h^{-1}; (b) 0, 6 km h^{-1}; (c) $\frac{15}{8}$ km h^{-1}, $\frac{45}{8}$ km h^{-1}

2.11 Because, for an arbitrarily small interval of time, the magnitude of displacement is equal to the length of the path.

2.12 All the four graphs are impossible. (a) a particle cannot have two different positions at the same time; (b) a particle cannot have velocity in opposite directions at the same time; (c) speed is always non-negative; (d) total path length of a particle can never decrease with time. (Note, the arrows on the graphs are meaningless).

2.13 No, wrong. x- t plot does not show the trajectory of a particle. Context: A body is dropped from a tower ($x = 0$) at $t = 0$.

2.14 105 m s^{-1}

2.15 (a) A ball at rest on a smooth floor is kicked, it rebounds from a wall with reduced speed and moves to the opposite wall which stops it; (b) A ball thrown up with some initial velocity rebounding from the floor with reduced speed after each hit; (c) A uniformly moving cricket ball turned back by hitting it with a bat for a very short time-interval.

2.16 $x < 0$, $v < 0$, $a > 0$; $x > 0$, $v > 0$, $a < 0$; $x < 0$, $v > 0$, $a > 0$.

2.17 Greatest in 3, least in 2; $v > 0$ in 1 and 2, $v < 0$ in 3.

2.18 Acceleration magnitude greatest in 2; speed greatest in 3; $v > 0$ in 1, 2 and 3; $a > 0$ in 1 and 3, $a < 0$ in 2; $a = 0$ at A, B, C, D.

Chapter 3

3.1 Volume, mass, speed, density, number of moles, angular frequency are scalars; the rest are vectors.

3.2 Work, current

3.3 Impulse

3.4 Only (c) and (d) are permissible

3.5 (a) T, (b) F, (c) F, (d) T, (e) T

3.6 Hint: The sum (difference) of any two sides of a triangle is never less (greater) than the third side. Equality holds for collinear vectors.

3.7 All statements except (a) are correct

3.8 400 m for each; B

3.9 (a) O; (b) O; (c) 21.4 km h^{-1}

3.10 Displacement of magnitude 1 km and direction 60° with the initial direction; total path length = 1.5 km (third turn); null displacement vector; path length = 3 km (sixth turn); 866 m, 30°, 4 km (eighth turn)

© NCERT not to be republished

3.11 (a) 49.3 km h^{-1} ; (b) 21.4 km h^{-1}. No, the average speed equals average velocity magnitude only for a straight path.

3.12 150.5 m

3.13 50 m

3.14 9.9 m s^{-2}, along the radius at every point towards the centre.

3.15 6.4 g

3.16 (a) False (true only for uniform circular motion)

(b) True, (c) True.

3.17 (a) $\mathbf{v}(t) = (3.0\,\hat{\mathbf{i}} - 4.0t\,\hat{\mathbf{j}})$ $\hat{\mathbf{a}}(t) = -4.0\hat{\mathbf{j}}$

(b) 8.54 m s^{-1}, 70° with x-axis.

3.18 (a) 2 s, 24 m, 21.26 m s^{-1}

3.19 $\sqrt{2}$, 45° with the x-axis; $\sqrt{2}$, – 45° with the x - axis, $\left(5/\sqrt{2},\ -1/\sqrt{2}\right)$.

3.20 (b) and (e)

3.21 Only (e) is true

3.22 182 m s^{-1}

Chapter 4

4.1 (a) to (d) No net force according to the First Law
(e) No force, since it is far away from all material agencies producing electromagnetic and gravitational forces.

4.2 The only force in each case is the force of gravity, (neglecting effects of air) equal to 0.5 N vertically downward. The answers do not change, even if the motion of the pebble is not along the vertical. The pebble is not at rest at the highest point. It has a constant horizontal component of velocity throughout its motion.

4.3 (a) 1 N vertically downwards (b) same as in (a)
(c) same as in (a); force at an instant depends on the situation at that instant, not on history.
(d) 0.1 N in the direction of motion of the train.

4.4 (i) T

4.5 $a = -2.5$ m s^{-2}. Using $v = u + at$, $0 = 15 - 2.5\,t$ i.e., $t = 6.0$ s

4.6 $a = 1.5/25 = 0.06$ m s^{-2}
$F = 3 \times 0.06 = 0.18$ N in the direction of motion.

4.7 Resultant force = 10 N at an angle of $\tan^{-1}(3/4) = 37°$ with the direction of 8 N force. Acceleration = 2 m s^{-2} in the direction of the resultant force.

4.8 $a = -2.5$ m s^{-2}, Retarding force = $465 \times 2.5 = 1.2 \times 10^3$ N

4.9 $F - 20{,}000 \times 10 = 20000 \times 5.0$, i.e., $F = 3.0 \times 10^5$ N

4.10 $a = -20$ m s^{-2} $0 \le t \le 30$ s
$t = -5$ s : $x = u\,t = -10 \times 5 = -50$ m

© NCERT not to be republished

$t = 25$ s : $x = ut + (½)\ at^2 = (10 \times 25 - 10 \times 625)$m $= -6$ km

$t = 100$ s : First consider motion up to 30 s

$x_1 = 10 \times 30 - 10 \times 900 = -8700$ m

At $t = 30$ s, $v = 10 - 20 \times 30 = -590$ m s^{-1}

For motion from 30 s to 100 s : $x_2 = -590 \times 70 = -41300$ m

$x = x_1 + x_2 = -50$ km

4.11 (a) Velocity of car (at $t = 10$ s) = $0 + 2 \times 10 = 20$ m s^{-1}

By the First Law, the horizontal component of velocity is 20 m s^{-1} throughout.
Vertical component of velocity (at $t = 11$s) = $0 + 10 \times 1 = 10$ m s^{-1}

Velocity of stone (at $t = 11$s) $=\sqrt{20^2 + 10^2} = \sqrt{500} = 22.4$ m s^{-1} at an angle of $\tan^{-1}$ (½) with the horizontal.
(b) 10 m s^{-2} vertically downwards.

4.12 (a) At the extreme position, the speed of the bob is zero. If the string is cut, it will fall vertically downwards.
(b) At the mean position, the bob has a horizontal velocity. If the string is cut, it will fall along a parabolic path.

4.13 The reading on the scale is a measure of the force on the floor by the man. By the Third Law, this is equal and opposite to the normal force N on the man by the floor.

(a) $N = 70 \times 10 = 700$ N ; Reading is 70 kg
(b) $70 \times 10 - N = 70 \times 5$; Reading is 35 kg
(c) $N - 70 \times 10 = 70 \times 5$; Reading is 105 kg
(d) $70 \times 10 - N = 70 \times 10$; Reading would be zero; the scale would read zero.

4.14 (a) In all the three intervals, acceleration and, therefore, force are zero.
(b) 3 kg m s^{-1} at $t = 0$; (c) –3 kg m s^{-1} at $t = 4$ s.

4.15 If the 20 kg mass is pulled,

$600 - T = 20\ a$, $T = 10\ a$

$a = 20$ m s^{-2}, $T = 200$ N

If the 10 kg mass is pulled, a = 20 m s^{-2}, $T = 400$ N

4.16 $T - 8 \times 10 = 8\ a$, $12 \times 10 - T = 12a$

i.e. $a = 2$ m s^{-2}, $T = 96$ N

4.17 By momentum conservation principle, total final momentum is zero. Two momentum vectors cannot sum to a null momentum unless they are equal and opposite.

4.18 Impulse on each ball = $0.05 \times 12 = 0.6$ kg m s^{-1} in magnitude. The two impulses are opposite in direction.

4.19 Use momentum conservation : $100\ v = 0.02 \times 80$

$v = 0.016$ m s^{-1} = 1.6 cm s^{-1}

4.20 Impulse is directed along the bisector of the initial and final directions. Its magnitude is $0.15 \times 2 \times 15 \times \cos 22.5° = 4.2$ kg m s^{-1}

4.21 $$v = 2\pi \times 1.5 \times \frac{40}{60} = 2\pi \text{ m s}^{-1}$$

$$T = \frac{mv^2}{R} = \frac{0.25 \times 4\pi^2}{1.5} = 6.6 \text{ N}$$

$$200 = \frac{mv_{max}^2}{R}, \text{ which gives } v_{max} = 35 \text{ m s}^{-1}$$

© NCERT not to be republished

4.22 Alternative (b) is correct, according to the First Law

4.23 (a) The horse-cart system has no external force in empty space. The mutual forces between the horse and the cart cancel (Third Law). On the ground, the contact force between the system and the ground (friction) causes their motion from rest.

(b) Due to inertia of the body not directly in contact with the seat.
(c) A lawn mower is pulled or pushed by applying force at an angle. When you push, the normal force (N) must be more than its weight, for equilibrium in the vertical direction. This results in greater friction f ($f \propto N$) and, therefore, a greater applied force to move. Just the opposite happens while pulling.

(d) To reduce the rate of change of momentum and hence to reduce the force necessary to stop the ball.

Chapter 5

5.1 (a) +ve (b) –ve (c) –ve (d) + ve (e) – ve

5.2 (a) 882 J ; (b) –247 J; (c) 635 J ; (d) 635 J;
Work done by the net force on a body equals change in its kinetic energy.

5.3 (a) $x > a$; 0 (c) $x < a$, $x > b$; $-V_1$
(b) $-\infty < x < \infty$; V_1 (d) $-b/2 < x < -a/2$, $a/2 < x < b/2$; $-V_1$

5.5 (a) rocket; (b) For a conservative force work done over a path is minus of change in potential energy. Over a complete orbit, there is no change in potential energy; (c) K.E. increases, but P.E. decreases, and the sum decreases due to dissipation against friction; (d) in the second case.

5.6 (a) decrease; (b) kinetic energy; (c) external force; (d) total linear momentum, and also total energy (if the system of two bodies is isolated).

5.7 (a) F ; (b) F ; (c) F ; (d) F (true usually but not always, why?)

5.8 (a) No
(b) Yes
(c) Linear momentum is conserved during an inelastic collision, kinetic energy is, of course, not conserved even after the collision is over.
(d) elastic.

5.9 (b) t

5.10 (c) $t^{3/2}$

5.11 12 J

5.12 The electron is faster, $v_e / v_p = 13.5$

5.13 0.082 J in each half ; – 0.163 J

5.14 Yes, momentum of the molecule + wall system is conserved. The wall has a recoil momentum such that the momentum of the wall + momentum of the outgoing molecule equals momentum of the incoming molecule, assuming the wall to be stationary initially. However, the recoil momentum produces negligible velocity because of the large mass of the wall. Since kinetic energy is also conserved, the collision is elastic.

5.15 43.6 kW

5.16 (b)

5.17 It transfers its entire momentum to the ball on the table, and does not rise at all.

5.18 5.3 m s^{-1}

© NCERT not to be republished

5.19 27 km h^{-1} (no change in speed)

5.20 50 J

5.21 (a) $m = \rho Avt$ (b) $K = \rho Av^3 t/2$ (c) $P = 4.5$ kW

5.22 (a) 49,000 J (b) 6.45 10^{-3} kg

5.23 (a) 200 m^2 (b) comparable to the roof of a large house of dimension 14m 14m.

Chapter 6

6.1 The geometrical centre of each. No, the CM may lie outside the body, as in case of a ring, a hollow sphere, a hollow cylinder, a hollow cube etc.

6.2 Located on the line joining H and C1 nuclei at a distance of 1.24 Å from the H end.

6.3 The speed of the CM of the (trolley + child) system remains unchanged (equal to v) because no external force acts on the system. The forces involved in running on the trolley are internal to this system.

6.6 $l_z = xp_y - yp_x$, $l_x = yp_z - zp_y$, $l_y = zp_x - xp_z$

6.8 72 cm

6.9 3675 N on each front wheel, 5145 N on each back wheel.

6.10 Sphere

6.11 Kinetic Energy = 3125 J; Angular Momentum = 62.5 J s

6.12 (a) 100 rev/min (use angular momentum conservation).

(b) The new kinetic energy is 2.5 times the initial kinetic energy of rotation. The child uses his internal energy to increase his rotational kinetic energy.

6.13 25 s^{-2}; 10 m s^{-2}

6.14 36 kW

6.15 at R/6 from the center of original disc opposite to the center of cut portion.

6.16 66.0 g

6.17 6.75 10^{12} rad s^{-1}

Chapter 7

7.1 (a) No.
(b) Yes, if the size of the space ship is large enough for him to detect the variation in g.
(c) Tidal effect depends inversely on the cube of the distance unlike force, which depends inversely on the square of the distance.

7.2 (a) decreases; (b) decreases; (c) mass of the body; (d) more.

7.3 Smaller by a factor of 0.63.

7.5 3.54 10^8 years.

7.6 (a) Kinetic energy, (b) less,

7.7 (a) No, (b) No, (c) No, (d) Yes

[The escape velocity is independent of mass of the body and the direction of projection. It depends upon the gravitational potential at the point from where the body is launched. Since this potential depends (slightly) on the latitude and height of the point, the escape

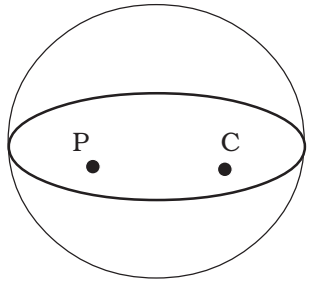
© NCERT not to be republished

velocity (speed) depends (slightly) on these factors.]

7.8 All quantities vary over an orbit except angular momentum and total energy.

7.9 (b), (c) and (d)

7.10 and **7.11** For these two problems, complete the hemisphere to sphere. At both P, and C, potential is constant and hence intensity = 0. Therefore, for the hemisphere, (c) and (e) are correct.



7.12 2.6×10^{8} m

7.13 2.0×10^{30} kg

7.14 1.43×10^{12} m

7.15 28 N

7.16 125 N

7.17 8.0×10^{6} m from the earth's centre

7.18 31.7 km/s

7.19 5.9×10^{9} J

7.20 2.6×10^{6} m/s

7.21 0, 2.7×10^{-8} J/kg; an object placed at the mid point is in an unstable equilibrium

© NCERT not to be republished

© NCERT not to be republished

© NCERT
be republished

ANSWERS

Chapter 8

8.1 1.8

8.2 (a) From the given graph for a stress of 150×10^6 N m^{-2} the strain is 0.002

(b) Approximate yield strength of the material is 3×10^8 N m^{-2}

8.3 (a) Material A

(b) Strength of a material is determined by the amount of stress required to cause fracture: material A is stronger than material B.

8.4 (a) False (b) True

8.5 1.5×10^{-4} m (steel); 1.3×10^{-4} m (brass)

8.6 Deflection = 4×10^{-6} m

8.7 2.8×10^{-6}

8.8 0.127

8.9 7.07×10^4 N

8.10 $D_{copper}/D_{iron} = 1.25$

8.11 1.539×10^{-4} m

8.12 2.026×10^9 Pa

8.13 1.034×10^3 kg/m^3

8.14 0.0027

8.15 0.058 cm^3

8.16 2.2×10^6 N/m^2

© NCERT not to be republished

Chapter 9

9.3 (a) decreases (b) η of gases increases, η of liquid decreases with temperature (c) shear strain, rate of shear strain (d) conservation of mass, Bernoulli's equation (e) greater.

9.5 6.2×10^6 Pa

9.6 10.5 m

19.7 Pressure at that depth in the sea is about 3×10^7 Pa. The structure is suitable since it can withstand far greater pressure or stress.

9.8 6.92×10^5 Pa

9.9 0.800

9.10 Mercury will rise in the arm containing spirit; the difference in levels of mercury will be 0.221 cm.

9.11 No, Bernoulli's principle applies to streamline flow only.

9.12 No, unless the atmospheric pressures at the two points where Bernoulli's equation is applied are significantly different.

9.13 9.8×10^2 Pa (The Reynolds number is about 0.3 so the flow is laminar).

9.14 1.5×10^3 N

9.15 Fig (a) is incorrect [Reason: at a constriction (i.e. where the area of cross-section of the tube is smaller), flow speed is larger due to mass conservation. Consequently pressure there is smaller according to Bernoulli's equation. We assume the fluid to be incompressible].

9.16 0.64 m s^{-1}

9.17 2.5×10^{-2} N m^{-1}

9.18 4.5×10^{-2} N for (b) and (c), the same as in (a).

9.19 Excess pressure = 310 Pa, total pressure = 1.0131×10^5 Pa. However, since data are correct to three significant figures, we should write total pressure inside the drop as 1.01×10^5 Pa.

9.20 Excess pressure inside the soap bubble = 20.0 Pa; excess pressure inside the air bubble in soap solution = 10.0 Pa. Outside pressure for air bubble = $1.01 \times 10^5 + 0.4 \times 10^3 \times 9.8 \times 1.2 = 1.06 \times 10^5$ Pa. The excess pressure is so small that up to three significant figures, total pressure inside the air bubble is 1.06×10^5 Pa.

Chapter 10

10.1 Neon: – 248.58 °C = – 415.44 °F;

CO_2: – 56.60 °C = – 69.88 °F

(use $t_F = \frac{9}{5} t_c + 32$)

© NCERT not to be republished

10.2 $T_A = (4/7)\ T_B$

10.3 384.8 K

10.4 (a) Triple-point has a *unique* temperature; fusion point and boiling point temperatures depend on pressure; (b) The other fixed point is the absolute zero itself; (c) Triple-point is 0.01°C, not 0 °C; (d) 491.69.

10.5 (a) T_A = 392.69 K, T_B = 391.98 K; (b) The discrepancy arises because the gases are not perfectly ideal. To reduce the discrepancy, readings should be taken for lower and lower pressures and the plot between temperature measured versus absolute pressure of the gas at triple point should be extrapolated to obtain temperature in the limit pressure tends to zero, when the gases approach ideal gas behaviour.

10.6 Actual length of the rod at 45.0 °C = (63.0 + 0.0136) cm = 63.0136 cm. (However, we should say that change in length up to three significant figures is 0.0136 cm, but the total length is 63.0 cm, up to three significant places. Length of the same rod at 27.0 °C = 63.0 cm.

10.7 When the shaft is cooled to temperature – 69^{0}C the wheel can slip on the shaft.

10.8 The diameter increases by an amount = 1.44×10^{-2} cm.

10.9 3.8×10^2 N

10.10 Since the ends of the combined rod are not clamped, each rod expands freely.

Δl_{brass} = 0.21 cm, Δl_{steel} = 0.126 cm = 0.13 cm

Total change in length = 0.34 cm. No 'thermal stress' is developed at the junction since the rods freely expand.

10.11 0.0147 = 1.5×10^{-2}

10.12 103 °C

10.13 1.5 kg

10.14 0.43 J g^{-1} K^{-1} ; smaller

10.15 The gases are diatomic, and have other degrees of freedom (i.e. have other modes of motion) possible besides the translational degrees of freedom. To raise the temperature of the gas by a certain amount, heat is to be supplied to increase the average energy of all the modes. Consequently, molar specific heat of diatomic gases is more than that of monatomic gases. It can be shown that if only rotational modes of motion are considered, the molar specific heat of diatomic gases is nearly (5/2) R which agrees with the observations for all the gases listed in the table, except chlorine. The higher value of molar specific heat of chlorine indicates that besides rotational modes, vibrational modes are also present in chlorine at room temperature.

10.16 4.3 g/min

10.17 3.7 kg

10.18 238 °C

10.20 9 min

© NCERT not to be republished

Chapter 11

11.1 16 g per min

11.2 934 J

11.4 2.64

11.5 16.9 J

11.6 (a) 0.5 atm (b) zero (c) zero (assuming the gas to be ideal) (d) No, since the process (called free expansion) is rapid and cannot be controlled. The intermediate states are non-equilibrium states and do not satisfy the gas equation. In due course, the gas does return to an equilibrium state.

11.7 25 W

11.8 450 J

Chapter 12

12.1 4×10^{-4}

12.3 (a) The dotted plot corresponds to 'ideal' gas behaviour; (b) $T_1 > T_2$; (c) 0.26 J K^{-1}; (d) No, 6.3×10^{-5} kg of H_2 would yield the same value

12.4 0.14 kg

12.5 5.3×10^{-6} m^3

12.6 6.10×10^{26}

12.7 (a) 6.2×10^{-21} J (b) 1.24×10^{-19} J (c) 2.1×10^{-16} J

12.8 Yes, according to Avogadro's law. No, v_{rms} is largest for the lightest of the three gases; neon.

12.9 2.52×10^3 K

12.10 Use the formula for mean free path :

$$\bar{l} = \frac{1}{\sqrt{2}\pi n d^2}$$

where d is the diameter of a molecule. For the given pressure and temperature $N/V = 5.10 \times 10^{25}$ m^{-3} and $= 1.0 \times 10^{-7}$ m. $v_{rms} = 5.1 \times 10^2$ m s^{-1}.

© NCERT not to be republished

collisional frequency = $\frac{v_{rms}}{\bar{l}}$ = $5.1 \times 10^9\ s^{-1}$. Time taken for the collision = d / v_{rms} = 4×10^{-13} s.

Time taken between successive collisions = l / v_{rms} = 2×10^{-10} s. Thus the time taken between successive collisions is 500 times the time taken for a collision. Thus a molecule in a gas moves essentially free for most of the time.

Chapter 13

13.1 (b), (c)

13.2 (b) and (c): SHM; (a) and (d) represent periodic but not SHM [A polyatomic molecule has a number of natural frequencies; so in general, its vibration is a superposition of SHM's of a number of different frequencies. This superposition is periodic but not SHM].

13.3 (b) and (d) are periodic, each with a period of 2 s; (a) and (c) are not periodic. [Note in (c), repetition of merely one position is not enough for motion to be periodic; the entire motion during one period must be repeated successively].

13.4 (a) Simple harmonic, $T = (2\pi/\omega)$; (b) periodic, $T = (2\pi/\omega)$ but not simple harmonic; (c) simple harmonic, $T = (\pi/\omega)$; (d) periodic, $T = (2\pi/\omega)$ but not simple harmonic; (e) non-periodic; (f) non-periodic (physically not acceptable as the function $\to \infty$ as $t \to \infty$.

13.5 (a) 0, +, + ; (b) 0, –, – ; (c) –, 0,0 ; (d) –, –, – ; (e) +, +, + ; (f) –, –, –.

13.6 (c) represents a simple harmonic motion.

13.7 A = $\sqrt{2}$ cm, $\phi = 7\pi/4$; B = $\sqrt{2}$ cm, $a = \pi/4$.

13.8 219 N

13.9 Frequency 3.2 s^{-1}; maximum acceleration of the mass 8.0 m s^{-2}; maximum speed of the mass 0.4 m s^{-1}.

13.10 (a) $x = 2 \sin 20t$
(b) $x = 2 \cos 20t$
(c) $x = -2 \cos 20t$

where x is in cm. These functions differ neither in amplitude nor frequency. They differ in initial phase.

13.11 (a) $x = -3 \sin \pi t$ where x is in cm.

(b) $x = -2 \cos \frac{\pi}{2} t$ where x is in cm.

13.13 (a) F/k for both (a) and (b).

(b) $T = 2\pi\sqrt{\frac{m}{k}}$ for (a) and $2\pi\sqrt{\frac{m}{2k}}$ for (b)

© NCERT not to be republished

13.14 100 m/min

13.15 8.4 s

13.16 $T = 2\pi\sqrt{\dfrac{l}{\sqrt{g^2 + v^4/R^2}}}$. Hint: Effective acceleration due to gravity will get reduced due to radial acceleration v^2/R acting in the horizontal plane.

13.17 In equilibrium, weight of the cork equals the up thrust. When the cork is depressed by an amount x, the net upward force is $Ax\rho_l g$. Thus the force constant $k = A\rho_l g$.

Using $m = Ah\rho$, and $T = 2\pi\sqrt{\dfrac{m}{k}}$ one gets the given expression.

13.18 When both the ends are open to the atmosphere, and the difference in levels of the liquid in the two arms is h, the net force on the liquid column is $Ah\rho g$ where A is the area of cross-section of the tube and ρ is the density of the liquid. Since restoring force is proportional to h, motion is simple harmonic.

Chapter 14

14.1 0.5 s

14.2 8.7 s

14.3 2.06×10^4 N

14.4 Assume ideal gas law: $P = \dfrac{\rho RT}{M}$, where ρ is the density, M is the molecular mass, and T is the temperature of the gas. This gives $v = \sqrt{\dfrac{\gamma RT}{M}}$. This shows that v is:

(a) Independent of pressure.

(b) Increases as $\sqrt{T}$.

(c) The molecular mass of water (18) is less than that of N_2 (28) and O_2 (32).

Therefore as humidity increases, the effective molecular mass of air decreases and hence v increases.

14.5 The converse is not true. An obvious requirement for an acceptable function for a travelling wave is that it should be finite everywhere and at all times. Only function (c) satisfies this condition, the remaining functions cannot possibly represent a travelling wave.

14.6 (a) 3.4×10^{-4} m (b) 1.49×10^{-3} m

© NCERT not to be republished

14.7 4.1×10^{-4} m

14.8 (a) A travelling wave. It travels from right to left with a speed of 20 ms^{-1}.

(b) 3.0 cm, 5.7 Hz

(c) $\pi/4$

(d) 3.5 m

14.9 All the graphs are sinusoidal. They have same amplitude and frequency, but different initial phases.

14.10 (a) 6.4 π rad

(b) 0.8 π rad

(c) π rad

(d) $(\pi/2)$ rad

14.11 (a) Stationary wave

(b) $l = 3$ m, n = 60 Hz, and $v = 180$ m s^{-1} for each wave

(c) 648 N

14.12 (a) All the points except the nodes on the string have the same frequency and phase, but not the same amplitude.

(b) 0.042 m

14.13 (a) Stationary wave.

(b) Unacceptable function for any wave.

(c) Travelling harmonic wave.

(d) Superposition of two stationary waves.

14.14 (a) 79 m s^{-1}

(b) 248 N

14.15 347 m s^{-1}

Hint : $v_n = \dfrac{(2n-1)v}{4l}$; $n = 1,2,3,....$for a pipe with one end closed

14.16 5.06 km s^{-1}

14.17 First harmonic (fundamental); No.

14.18 318 Hz

© NCERT not to be republished

BIBLIOGRAPHY

TEXTBOOKS

For additional reading on the topics covered in this book, you may like to consult one or more of the following books. Some of these books however are more advanced and contain many more topics than this book.

1. **Ordinary Level Physics**, A.F. Abbott, Arnold-Heinemann (1984).
2. **Advanced Level Physics**, M. Nelkon and P. Parker, 6th Edition Arnold-Heinemann (1987).
3. **Advanced Physics**, Tom Duncan, John Murray (2000).
4. **Fundamentals of Physics**, David Halliday, Robert Resnick and Jearl Walker, 7th Edition John Wily (2004).
5. **University Physics**, H.D. Young, M.W. Zemansky and F.W. Sears, Narosa Pub. House (1982).
6. **Problems in Elementary Physics**, B. Bukhovtsa, V. Krivchenkov, G. Myakishev and V. Shalnov, MIR Publishers, (1971).
7. **Lectures on Physics** (3 volumes), R.P. Feynman, Addision – Wesley (1965).
8. **Berkeley Physics Course** (5 volumes) McGraw Hill (1965).
   a. Vol. 1 – Mechanics: (Kittel, Knight and Ruderman)
   b. Vol. 2 – Electricity and Magnetism (E.M. Purcell)
   c. Vol. 3 – Waves and Oscillations (Frank S. Craw-ford)
   d. Vol. 4 – Quantum Physics (Wichmann)
   e. Vol. 5 – Statistical Physics (F. Reif)
9. **Fundamental University Physics**, M. Alonso and E. J. Finn, Addison – Wesley (1967).
10. **College Physics**, R.L. Weber, K.V. Manning, M.W. White and G.A. Weygand, Tata McGraw Hill (1977).
11. **Physics**: **Foundations and Frontiers**, G. Gamow and J.M. Cleveland, Tata McGraw Hill (1978).
12. **Physics for the Inquiring Mind**, E.M. Rogers, Princeton University Press (1960)
13. **PSSC Physics Course**, DC Heath and Co. (1965) Indian Edition, NCERT (1967)
14. **Physics Advanced Level**, Jim Breithampt, Stanley Thornes Publishers (2000).
15. **Physics,** Patrick Fullick, Heinemann (2000).

© NCERT not to be republished

16. **Conceptual Physics,** Paul G. Hewitt, Addision-Wesley (1998).
17. **College Physics,** Raymond A. Serway and Jerry S. Faughn, Harcourt Brace and Co. (1999).
18. **University Physics,** Harris Benson, John Wiley (1996).
19. **University Physics,** William P. Crummet and Arthur B. Western, Wm.C. Brown (1994).
20. **General Physics,** Morton M. Sternheim and Joseph W. Kane, John Wiley (1988).
21. **Physics,** Hans C. Ohanian, W.W. Norton (1989).
22. **Advanced Physics,** Keith Gibbs, Cambridge University Press(1996).
23. **Understanding Basic Mechanics,** F. Reif, John Wiley (1995).
24. **College Physics,** Jerry D. Wilson and Anthony J. Buffa, Prentice-Hall (1997).
25. **Senior Physics, Part – I,** I.K. Kikoin and A.K. Kikoin, Mir Publishers (1987).
26. **Senior Physics, Part – II,** B. Bekhovtsev, Mir Publishers (1988).
27. **Understanding Physics,** K. Cummings, Patrick J. Cooney, Priscilla W. Laws and Edward F. Redish, John Wiley (2005)
28. **Essentials of Physics,** John D. Cutnell and Kenneth W. Johnson, John Wiley (2005)

GENERAL BOOKS

For instructive and entertaining general reading on science, you may like to read some of the following books. Remember however, that many of these books are written at a level far beyond the level of the present book.

1. **Mr. Tompkins** in paperback, G. Gamow, Cambridge University Press (1967).
2. **The Universe and Dr. Einstein**, C. Barnett, Time Inc. New York (1962).
3. **Thirty years that Shook Physics**, G. Gamow, Double Day, New York (1966).
4. **Surely You're Joking, Mr. Feynman**, R.P. Feynman, Bantam books (1986).
5. **One, Two, Three... Infinity**, G. Gamow, Viking Inc. (1961).
6. **The Meaning of Relativity**, A. Einstein, (Indian Edition) Oxford and IBH Pub. Co (1965).
7. **Atomic Theory and the Description of Nature**, Niels Bohr, Cambridge (1934).
8. **The Physical Principles of Quantum Theory**, W. Heisenberg, University of Chicago Press (1930).
9. **The Physics- Astronomy Frontier**, F. Hoyle and J.V. Narlikar, W.H. Freeman (1980).
10. **The Flying Circus of Physics with Answer**, J. Walker, John Wiley and Sons (1977).
11. **Physics for Everyone** (series), L.D. Landau and A.I. Kitaigorodski, MIR Publisher (1978).
    Book 1: Physical Bodies
    Book 2: Molecules
    Book 3: Electrons
    Book 4: Photons and Nuclei.
12. **Physics can be Fun**, Y. Perelman, MIR Publishers (1986).
13. **Power of Ten**, Philip Morrison and Eames, W.H. Freeman (1985).
14. **Physics in your Kitchen Lab**., I.K. Kikoin, MIR Publishers (1985).
15. **How Things Work : The Physics of Everyday Life**, Louis A. Bloomfield, John Wiley (2005)
16. **Physics Matters : An Introduction to Conceptual Physics,** James Trefil and Robert M. Hazen, John Wiley (2004).

© NCERT not to be republished

INDEX

A

Absolute scale temperature 280
Absolute zero 280
Acceleration (linear) 45
Acceleration due to gravity 49,189
Accuracy 22
Action-reaction 97
Addition of vectors 67
Adiabatic process 311, 312
Aerofoil 262
Air resistance 79
Amplitude 344, 372
Angle of contact 267, 268
Angstrom 21
Angular Acceleration 154
Angular displacement 342
Angular frequency 344, 373
Angular momentum 155
Angular velocity 152
Angular wave number 372
Antinodes 381,382
Archimedes Principle 255
Area expansion 281
Atmospheric pressure 253
Average acceleration 45, 74
Average speed 42
Average velocity 42
Avogardo's law 325

B

Banked road 104
Barometer 254
Beat frequency 383
Beats 382, 383
Bending of beam 244
Bernoulli's Principle 258
Blood pressure 276
Boiling point 287
Boyle's law 326
Buckling 244
Bulk modulus 242
Buoyant force 255

C

Calorimeter 285
Capillary rise 268
Capillary waves 370
Carnot engine 316
Central forces 186
Centre of Gravity 161
Centre of mass 144
Centripetal acceleration 81
Centripetal force 104
Change of state 287
Charle's law 326
Chemical Energy 126
Circular motion 104
Clausius statement 315
Coefficient of area expansion 283
Coefficient of linear expansion 281
Coefficient of performance 314
Coefficient of static friction 101
Coefficient of viscosity 262
Coefficient of volume expansion 281
Cold reservoir 313
Collision 129
Collision in two dimensions 131
Compressibility 242, 243
Compressions 368, 369, 374
Compressive stress 236, 243
Conduction 290
Conservation laws 12
Conservation of angular momentum 157, 173
Conservation of Mechanical Energy 121
Conservation of momentum 98
Conservative force 121
Constant acceleration 46,75
Contact force 100
Convection 293
Couple 159
Crest 371
Cyclic process 312

© NCERT not to be republished

D

Dalton's law of partial pressure 325
Damped oscillations 355
Damped simple Harmonic motion 355
Damping constant 355
Damping force 355
Derived units 16
Detergent action 269
Diastolic pressure 277
Differential calculus 61
Dimensional analysis 32
Dimensions 31
Displacement vector 66
Displacement 40
Doppler effect 385, 386
Doppler shift 387
Driving frequency 358
Dynamics of rotational motion 169

E

Efficiency of heat engine 313
Elastic Collision 129
Elastic deformation 236, 238
Elastic limit 238
Elastic moduli 239
Elasticity 235
Elastomers 239
Electromagnetic force 8
Energy 117
Equality of vectors 66
Equation of continuity 257
Equilibrium of a particle 99
Equilibrium of Rigid body 158
Equilibrium position 341, 342, 353
Errors in measurement 22
Escape speed 193

F

First law of Thermodynamics 307
Fluid pressure 251
Force 94
Forced frequency 357
Forced oscillations 357, 358
Fracture point 238
Free Fall 49
Free-body diagram 100
Frequency of periodic motion 342, 372
Friction 101
Fundamental Forces 6
Fundamental mode 381
Fusion 287

G

Gauge pressure 253
Geocentric model 183
Geostationary satellite 196
Gravitational constant 189
Gravitational Force 8, 192
Gravitational potential energy 191
Gravity waves 370

H

Harmonic frequency 380, 381
Harmonics 380, 381
Heat capacity 284
Heat engines 313
Heat pumps 313
Heat 279
Heliocentric model 183
Hertz 343
Hooke's law 238
Horizontal range 78
Hot reservoir 313
Hydraulic brakes 255, 256
Hydraulic lift 255, 256
Hydraulic machines 255
Hydraulic pressure 238
Hydraulic stress 238, 243
Hydrostatic paradox 253

I

Ideal gas equation 280
Ideal gas 280, 325
Impulse 96
Inelastic collision 129
Initial phase angle 372
Instantaneous acceleration 74
Instantaneous speed 45
Instantaneous velocity 43
Interference 377
Internal energy 306, 330
Irreversible engine 315, 317
Irreversible processes 315
Isobaric process 311, 312
Isochoric process 311, 312
Isotherm 310
Isothermal process 311

K

Kelvin-Planck statement 315
Kepler's laws of planetary motion 184
Kinematics of Rotational Motion 167
Kinematics 39
Kinetic energy of rolling motion 174
Kinetic Energy 117
Kinetic interpretation of temperature 329
Kinetic theory of gases 328

L

Laminar flow 258, 264
Laplace correction 376

© NCERT not to be republished

Latent heat of fusion 290
Latent heat of vaporisation 290
Latent heat 289
Law of cosine 72
Law of equipartition of energy 332
Law of Inertia 90
Law of sine 72
Linear expansion 281
Linear harmonic oscillator 349, 351
Linear momentum 155
Longitudinal strain 236
Longitudinal strain 236, 239
Longitudinal stress 236
Longitudinal Wave 369, 376

M

Magnus effect 261
Manometer 254
Mass Energy Equivalence 126
Maximum height of projectile 78
Maxwell Distribution 331
Mean free path 324, 335
Measurement of length 18
Measurement of mass 21
Measurement of temperature 279
Measurement of time 22
Melting point 286
Modes 380
Modulus of elasticity 238
Modulus of rigidity 242
Molar specific heat capacity at constant pressure 284, 308
Molar specific heat capacity at constant volume 284, 308
Molar specific heat capacity 284
Molecular nature of matter 323
Moment of Inertia 163
Momentum 93
Motion in a plane 72
Multiplication of vectors 67
Musical instruments 384

N

Natural frequency 358
Newton's first law of motion 91
Newton's Law of cooling 295
Newton's law of gravitation 185
Newton's second law of motion 93
Newton's third law of motion 96
Newtons' formula for speed of sound 377
Nodes 381
Normal Modes 381, 382, 384
Note 384, 385
Nuclear Energy 126
Null vector 68

O

Odd harmonics 382
Orbital velocity/speed 194
Order of magnitude 28
Oscillations 342
Oscillatory motion 342

P

Parallax method 18
Parallelogram law of addition of vectors 66
Pascal's law 252
Path length 40
Path of projectile 78
Periodic force 358
Periodic motion 342
Periodic time 342
Permanent set 238
Phase angle 344
Phase constant 344
Pipe open at both ends 382
Pipe open at one end 381
Pitch 384
Plastic deformation 238
Plasticity 235
Polar satellite 196
Position vector and displacement 73
Potential energy of a spring 123
Potential energy 120
Power 128
Precession 143
Pressure gauge 253
Pressure of an ideal gas 328
Pressure 250
Principle of Conservation of Energy 128
Principle of moments 160
Progressive wave 373
Projectile motion 77
Projectile 77
Propagation constant 371
Pulse 369

Q

Quasi-static process 310, 311

R

Radiation 294
Radius of Gyration 164
Raman effect 11
Rarefactions 369
Ratio of specific heat capacities 334
Reaction time 51
Real gases 326
Rectilinear motion 39
Reductionism 2
Reflected wave 379
Reflection of waves 378

© NCERT not to be republished

Refracted wave 379
Refrigerator 313
Regelation 287
Relative velocity in two dimensions 76
Relative velocity 51
Resolution of vectors 69
Resonance 358
Restoring force 236, 350, 369
Reversible engine 316, 317
Reversible processes 315
Reynolds number 264
Rigid body 141
Rolling motion 173
Root mean square speed 329
Rotation 142

S

S.H.M. (Simple Harmonic Motion) 343
Scalar-product 114
Scalars 65
Scientific Method 1
Second law of Thermodynamics 314
Shear modulus 242
Shearing strain 237
Shearing stress 237, 243
SI units 16
Significant figures 27
Simple pendulum 343, 353
Soap bubbles 268
Sonography 387
Sound 375
Specific heat capacity of Solids 308, 335
Specific heat capacity of Gases 333, 334
Specific heat capacity of Water 335
Specific heat capacity 285, 308
Speed of efflux 259
Speed of Sound 375, 376
Speed of Transverse wave on a stretched string 375, 376
Sphygmomanometer 277
Spring constant 352, 355
Standing waves 380
Stationary waves 382
Steady flow 257
Stethoscope 281
Stokes' law 263
Stopping distance 50
Strain 236
Streamline flow 257, 258
Streamline 257, 258
Stress 236
Stress-strain curve 238
Stretched string 374
Sublimation 294
Subtraction of vectors 67
Superposition principle 378
Surface energy 265
Surface tension 265
Symmetry 146
System of units 16
Systolic pressure 277

T

Temperature 279
Tensile strength 238
Tensile stress 236
Terminal velocity 264
Theorem of parallel axes 167
Theorem of perpendicular axes 165
Thermal conductivity 291
Thermal equilibrium 304
Thermal expansion 281
Thermal stress 284
Thermodynamic processes 310
Thermodynamic state variables 309
Thermodynamics 3, 303
Time of flight 78
Torque 154
Torricelli's Law 259, 260
Trade wind 294
Transmitted wave 379
Travelling wave 380
Triangle law of addition of vectors 66
Triple point 288
Trough 371
Tune 384
Turbulent flow 258, 259

U

Ultimate strength 238
Ultrasonic waves 387
Unification of Forces 10
Unified Atomic Mass Unit 21
Uniform circular motion 79
Uniform Motion 41
Uniformly accelerated motion 47
Unit vectors 70

V

Vane 356
Vaporisation 288
Vector-product 151
Vectors 66
Velocity amplitude 349
Venturi meter 260
Vibration 341
Viscosity 262
Volume expansion 281
Volume Strain 238

W

Wave equation 374
Wavelength 372
Wave speed 374

© NCERT not to be republished

Waves 368
Waxing and waning of sound 385
Weak nuclear force 9
Weightlessness 197
Work done by variable force 118
Work 116
Work-Energy Theorem 116
Working substance 313

Y

Yield Point 238
Yield strength 238
Young's modulus 239

Z

Zeroth law of Thermodynamics 305

© NCERT not to be republished

NOTES

© NCERT not to be republished